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Internet of Things and smart sensors in agriculture: Scopes and challenges

Prem Rajak [a,*], Abhratanu Ganguly [a], Satadal Adhikary [b], Suchandra Bhattacharya [c]

[a] Department of Animal Science, Kazi Nazrul University, Asansol, West Bengal, India
[b] Post Graduate Department of Zoology, A. B. N. Seal College, Cooch Behar, West Bengal, India
[c] Department of Chemistry, A. B. N. Seal College, Cooch Behar, West Bengal, India

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ABSTRACT

Agriculture is an essential sector needed for survival of the human community. Several measures have been taken to enhance the crop production. However harsh environmental conditions and frequent pest infestation lead to the agricultural loss. In such scenario, integration of advanced technologies such as advanced sensors coupled with Internet of Things (IoT) could escalate the agricultural production and minimize the economic loss. Studies have been conducted across the world that satisfactorily demonstrated the implication of integrated IoT-smart sensors in monitoring environmental factors such as moisture, humidity, temperature, and soil composition that are critical for crop growth. Green house gases such as Carbon dioxide, Methane, etc., are also measured through automated sensors. Smart farming also enables measurement of nitrogen contents in soil that helps farmers to determine the amount of fertilizers to be used in farm lands. Some IoT-enabled equipments and unmanned aerial vehicles are useful in accurate surveillance of pest attack and associated diseases in farm vegetation. Though the smart farming has great scopes in future, it faces certain limitations related to high implementation cost, data security, and lack of sufficient digital knowledge in farmers. Special economic policies, data encryption, and digital literacy could ease IoT-enabled smart farming in future.

1. Introduction

Artificial Intelligence (AI) is one of the fastest growing technologies in this modern world. AI is now extensively employed in various industries ranging from the development of robotics to the monitoring of environmental issues [1]. AI technology has undeniable scopes in health care devices, self driving vehicles, satellite imaging, landscape mapping, climate change monitoring, and health care systems [2]. AI has escalated the development of robotics across the globe that has made the life quite easier and contributed to a much better life of well being. AI mediated automation of several sectors including agriculture will generate good quality products with less negative impacts on environment [3]. AI employs a variety of smart sensors that process data in real time with higher accuracy. Hence, AI has enhanced the performance of any device to several folds with greater sensibility and precision in their outcomes. AI with smart sensors has lead to the development of an advanced technology known as the Internet of Things (IoT). IoT is hub of wireless systems and AI deals with data in real time and generates appropriate outcomes. This technology has been adopted in almost every sector of science and research. IoT based digital technologies have potential to generate innovative models for better production of consumer products including agro-products [4].

Agriculture is integral part of the human community. It has been upgraded with time to increase agricultural yields. Modern technologies have also been employed in agriculture, horticulture, and forestry for better monitoring of plant growth, diseases, and pest population. AI and IoT have numerous burgeoning scopes in agriculture to solve various issues like soil conditions, weather status, pest detection, determination of harvesting time, and so on [5]. In addition AI and IoT offers excellent scopes for real time surveillance of temperature, humidity, pollution, water content, soil quality, radiation, and others. Smart farming contributes to precision by better management of farm works and prompt decision making based on acquired data [6]. IoT coupled with various smart sensors assist the farmers for better management of crops and vegetables in a larger area and brief time. In addition, IoT could also facilitate to understand the need and time of pesticide as well as fertilizer application in the field. Excessive pesticide and fertilizer usage can lead to environmental contamination and health hazards [7–12]. Implication of IoT could certainly reduce the wastage of various resources as well as save the revenues of farmers. Moreover IoT and smart

* Corresponding author.
E-mail addresses: prem.rajak@knu.ac.in (P. Rajak), gangulyabhratanu@gmail.com (A. Ganguly), sadhikaryrnj@gmail.com (S. Adhikary), suchandrab7@gmail.com (S. Bhattacharya).

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sensors can be coupled with a variety of existing devices that could further enhance agricultural production by facilitating accurate diagnosis of crop diseases [13].

Precision farming employs IoT based technologies to monitor agricultural factors. IoT provides real time vital data related to the crop, soil, water, and air conditions for better protection of environment and continued sustainability of agricultural production. Irrigation can be transformed into the smart irrigation system to escalate the irrigation process in a short period of time. Equipments like unmanned aerial vehicles (UAVs) are used to collect important data regarding the critical factors of agriculture like pest attack, soil quality, and water availability in a large area within a short time [14]. UAVs integrated with spectrum analysis technology can capture high resolution images that are stored in cloud server databases for further analyses even from remote areas to detect pest infestation in farms [1]. Smart greenhouse has the emerging scope in cultivation process with least human intervention. It involves sensor based continuous surveillance of temperature, humidity, luminance, and soil moisture for better yield [15]. These automated sensors help take decision to implement more correct measures to avoid agricultural damage.

Farm management system (FMS) is an integrated strategy involving the coordination of IoT and AI. It offers better tracing system to analyze various physical, chemical, and biological agricultural factors to enhance both quality and quantity in agricultural production. It involves less manpower with real time data processing facilities. Therefore, FMS can help to maintain qualitative as well as sustainability of agricultural production with less effort and time.

IoT backed with an intelligent decision making system can significantly reduce the human intervention in management of agricultural sectors. The decision making system is central to all accessories that could be responsible for success or failure of the automated algorithm. IoT coupled with IT has potential to excel better than human decision in terms of accuracy and data processing. AI is a system that performs operations similar to human intelligences to sense speech recognition, visual perception, decision making, and automated data processing [16]. Indeed the IoT and AI are essential components of an integrated network with greater efficiency and accuracy in automated management of agriculture. IoT generates huge data that requires fast data analysis and such task can be completed through AI algorithms with better efficiency and high quality of decision making. New logic and methods like machine learning, machine vision, artificial neural networks, natural languages processing, etc., have improved the automation process in agriculture. Among these technologies, machine learning and artificial neural networks are the most widely used in research for automation of agriculture sector around the globe [17]. Machine learning algorithms work with supervised and unsupervised learning algorithms. Nonetheless, most of the automated systems follow the supervised algorithms. Artificial neural networks are inspired from architecture of human neurons organized in several layers. Artificial neural networks are devised to learn non-linear relationships. Conventional neural networks are now linked to deep learning-based computer vision techniques that are being widely researched for automation of agriculture. AI models are initially designed with known sets of data and tested for their accuracy before their deployment. AI can generate data and based on such data, smart decisions and recommendations are communicated to the farmers through SMS in their smart phones so that farmers can initiate the necessary actions. Optimal mobile network-based convolutional neural network model (OMNCNN) is useful in detecting plant diseases through different stages such as preprocessing, segmentation, feature extraction, and classification [18]. Comparatively, OMNCNN model has shown promising results over existing state-of-art methods with the maximum precision of 0.985, recall of 0.9892, and accuracy of 0.987 along with additional parameters. Proximal soil sensors coupled with computer vision with machine learning algorithms provide quick, accurate, and nondestructive predictions for motoring soil moisture and organic matter contents. For instance, soil organic matter and soil moisture content were predicted using high resolution digital images coupled with computer vision algorithms. Results were compared with laboratory measured soil properties. Exponential Gaussian Process Regression and Cubist model displayed best performance for soil moisture content prediction, with coefficients of determination values of 0.84 and 0.86, and root-mean-square-error (RMSE) of 10.18% and 10.43%, respectively. For soil organic matter, Artificial Neural Network and Cubist model generated satisfactory prediction accuracy with coefficients of determination values of 0.91 and 0.72, and RMSE values of 5.45% and 9.90%, respectively [19]. An algorithm employing Haar Wavelet Transform (HWT) and the classifier k-NN was devised to screen weeds with better efficiency and accuracy. The proposed method enhanced the working performance by 7–10% as compared with some of the existing techniques that used histogram maxima with threshold, and angular cross sectional intensities. Furthermore, results indicated the overall accuracy and efficiency of the proposed method utilizing the HWT as 94% and 40 ms respectively [20]. Deep learning frameworks like convolutional neural networks (CNNs) and recurrent neural networks (RNNs) model, random forest (RF), deep fully connected neural networks (DFNN), and LASSO are popularly used to predict crop yields. In a study, CNN-RNN model achieved an RMSE of 9% and 8% average yields of corn and soybean respectively in the United States for years 2016, 2017, and 2018, substantially outperforming other tested methods such as RF, DFNN, and LASSO [21]. Therefore, AI has promising potentials to enhances the efficacy and accuracy of IoTs employed in agriculture.

The present article aims to discuss the potential implication of IoT and smart sensors in precise monitoring of several physico-chemical factors like humidity, temperature, moisture contents, nitrogen contents, soil quality, and green house gases in agricultural lands. In addition, implication of smart traps equipped with high resolution cameras will also be presented. Final section of the paper will focus on the current challenges and possible solutions for deployment of IoT-based devices on large scale.

2. Remote sensing using smart sensors

Owing to the rapid developments in communication technologies, semiconductor industry, and computing, a new generation of sensors called “smart” sensors have originated as cutting-edge technologies which are capable of wireless communication from a remote site. These are capable of inter-device communication, along with automated data processing network. Thus “smart” sensors are very effective in the detection of regularities and correlations in raw data, and the establishment of cause and effect relationships among various factors. Earlier generations of sensor networks had only the sensing features while the later versions were boosted with intelligence to perform computations along with sensing data. What has made a sensor “smart” is enabling the sensors to facilitate communication among these intelligent sensing devices, thereby establishing a “smart sensor network”. The basic architecture of a smart sensor include one or more sensors, a microcontroller, an analog-to-digital converter (ADC), a communication link, memory, and a power source. A smart sensor node is made up of three fundamental components viz. a physical transducer, a memory core or processor, and a network interface. The physical parameters are sensed by the physical transducer, which are then converted into an electrical signal. The ADC aids in producing a digital value for use by the processor. Signal processing on the sensed data and conveying the resulting data to the network is done by the processor, which is often a microcontroller. Conventional sensor systems are made up of single central data processing resource. Following local processing of data at the signal node, the result is further communicated to the network [22].

Efficiency of remote sensing can be increased by tagging with the IoT enabled smart sensors that have potential to process and analyze real time data. Both active and passive sensors could be employed in remote sensing to motor various aspects of agriculture. Applicability of IoT

enabled smart sensors includes weather forecasting, soil monitoring, landscape topology, pest manifestation, and soil quality [23]. Smart sensors are based on AI technologies that enhance precision in their functions. LiDAR-based sensing technology offers precision agriculture by presenting real time and more accurate data. This technology provides real time information on soil quality, moisture content, crop growth, pest manifestation, growth status of vegetation, and productivity. It also offers generalized information on draught conditions and diseases of plants. IoT has implications in a wide range of tasks like data acquisition, processing, and analysis. Moreover, several environmental conditions like water contamination, pollution, soil quality assessment, radiation status, etc., are extensively studies using IoT technology [24–26]. In IoT enabled remote sensing, several processing, sensing, and signal transmissible units are connected together through a shared cloud for data sharing to the site of interest. Time series method for Sentinel-1 SAR data is useful to identify and classify various types of growth [27]. Hyper spectral imaging coupled with regression analysis method can be beneficial to meet up the various agricultural needs [28]. A model on remote sensing communication has potential implication in detection of wild fire in forestry in a time-sensitive manner [29]. Geospatial analysis tagged with IoT and environmental informatics is useful to interpret data related to the environmental research and agricultural applications [30].

In order to provide remote sensing capabilities for "smart environment" applications, sensors are also being tied to Radio Frequency IDentification (RFID) devices. An RFID system is made up of a reader, a tag, and an antenna. In this technology, radio communication between the tag and the reader aids in retrieving the stored data from the tag. Existing RFID technology with the integration of sensing elements is promising enormous possibilities in the remote sensing of temperature [31,32], humidity [33,34], and other environmental stimuli, which are crucial for agricultural practices.

3. IoT and agriculture

IoT, smart sensors, and AI have great scopes to acquire real time data and interpret them to monitor soil health, water contents, crop quality, and crop production in a site [35]. IoT coupled with smart sensors is replacing the conventional agricultural methods by smart farming characterized by enhanced yield. IoT enabled technological approaches in agriculture also helps to assess soil health, soil erosion, need of fertilizer, status of soil fertility, and crop quality [36]. It also supports seed quality, optical irrigation, and surveillance of crop growth at various stages. Notably, real data of remote sensing and IoT can be processed for precision agriculture and forestry. Approaches like infrared thermography coupled with smart sensors have been devised to sense topological data of an agricultural land. IoT and smart soil moisture sensors are used for preharvesting and postharvesting status in agricultural lands. Microbial entities impose adverse impacts on global population [37–39]. Microbial presence in agrifoods can be sensed by microfluidics-based on-chip artificial pore [40]. Onsite detection of pathogens generally comprises lateral flow tests and lateral flow assays [41]. Interestingly, deep learning approaches are used for interpretation of data acquired by multiple sensors. IoT, ZigBee, and Arduino sensors can be used for precise assessment of temperature, moisture, soil status, and crop types in a specified area [42]. Therefore data can be analyzed by the IoT for planned increment in agricultural and crop yields. IoT and UAVs are the major components of smart farming which can be managed at less man power with better yield. Therefore, smart farming can be improved by implication of machine-deep learning strategies and Arduino based controls [43]. Flying IoT provides more accurate real time data that further contribute to the smart farming. Evaluation of crop quality, drought control, ground water irrigation status, and soil health could have made easy by the cost effective platforms employing IoT. Some indices like Vegetation health index (VHI), Standardized anomaly index (SAI), and Evaporative stress index (ESI) are the major indices to promote smart farming. For automated intelligent governance of IoT functions, development and implementation of microprocessors have attracted global attention. Real time precise surveillance of field cultivation, water content, and fertilizer requirement in many developed countries are performed by retrieval of vital information from satellites and GPS system. Agricultural IoTs conjugated with expert systems can help farmers to enhance their planting and crop management strategies [44]. At present, the equipments developed and employed mostly include those for gathering agricultural and crop information, tracking environmental data, and monitoring animal movements [45]. Electromechanical sensors, biosensors, and physical property sensors have major implications in agricultural fields. Biological sensors are enabled to sense biological sensitive components from outside world however; physical property sensors employ devices sensitive to alterations in physical environment (Fig. 1).

As an example for smart farming, several groups put up an inventive IoT system using cloud computing and Li-Fi. Wireless data exposure with a high concentration in a confined space is known as Li-Fi. Compared to Wi-Fi, Li-Fi provides more bandwidth, efficiency, accessibility, and protection. The main purpose of this smart farming operation is to carry out tasks like spraying, weeding, moisture detecting, etc. By ensuring humidity protection and temperature preservation in the stockroom, it also embraces smart warehouse supervision. Finally, flawless irrigation with elegant management is essential for a clear and reliable execution of agricultural practices. Edging sensors, cameras, ZigBee, or Li-Fi modules carry out systematic smart mechanisms to accomplish all of these [46]. Since water is getting increasingly scarce today, prudential use of water bears enormous significance. A novel method known as Smart Irrigation Decision Support System (SIDSS) has shown great potential to efficiently manage water supply in agricultural lands [47]. Two machine learning approaches such as partial least squares regression (PLSR) and adaptive neuro-fuzzy inference system (ANFIS), have been proposed to achieve SIDSS. Three commercial citrus tree plantations in Spain have been employed to validate this methodology. In contrast to earlier studies that are based just on weather factors or do not indicate how much water is needed by the crops, the fundamental characteristic of the system is the use of continuous soil measurements to complement climatic parameters to correctly forecast the irrigation demands of the crops [48].

3.1. IoT-based crop monitoring

A paradigm shift in crop monitoring from an experience-based qualitative task into a quantitative and data-driven work has taken place with the introduction of IoT into soil, microclimate, and crop sensing. IoT based technology enables farmers to precisely monitor crop health and growth [5]. It also assists farm workers to assess pest attack and plant diseases in real time. Acquisition and processing of real time data by IoT tagged sensors offer great advantage to farmers and researchers for intelligent management of crop cultivation, fertilizer application, irrigation, and plant surroundings [49].

Sensors that are placed in the field to gather information on several environmental parameters, including temperature, humidity, soil moisture, and nutrient levels are the main components of IoT-based smart crop monitoring systems. These sensors are frequently networked, enabling real-time data transmission to a centralized cloud-based platform. To gain insights into crop health, growth rates, and yield potential, the data gathered from these sensors is then examined using machine learning and other data analytics techniques. Farmers can use this information to make choices on crop management including irrigation, pest control, and harvest.

Implication of wireless sensor technologies can alert administrators and farmers regarding equipment fault and initiates troubleshooting. Moreover automated repairing tool could help in reducing energy consumption, escalating data processing, and enhancing actuation [50]. IoT-enabled UAS are emerging trends in agriculture where the system

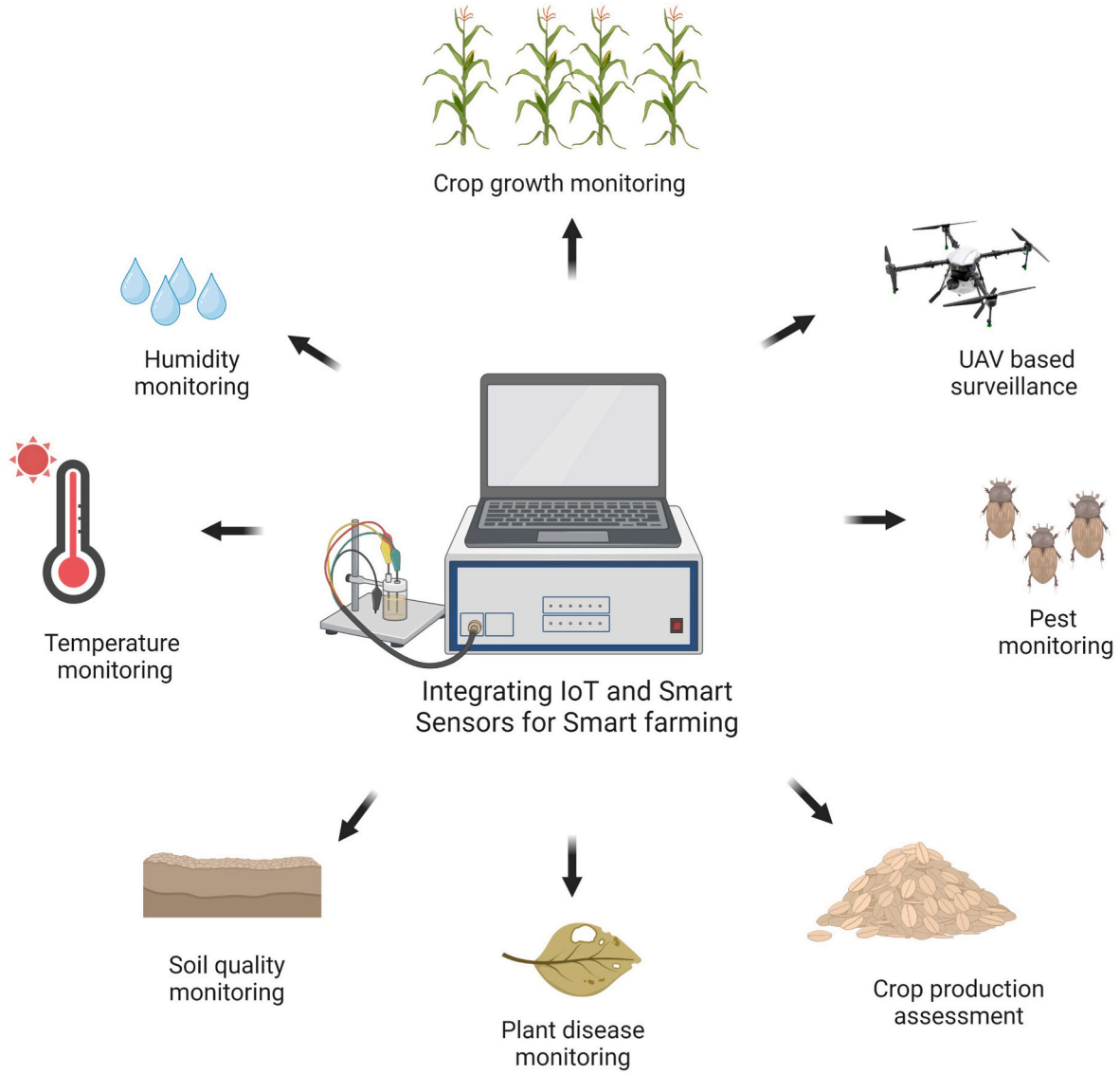

Fig. 1. Applications of integrated IoT and smart sensors for precision farming. IoT-based smart sensors can accurately monitor environmental factors such as temperature, moisture, and humidity. Some sensors can assess soil quality by determining nitrate levels and water content. Plant disease and insect pests can be detected using high resolution camera coupled with GPRS system. UAV based surveillance helps monitor crop growth and farm land topology. Crop production can be estimated by automated mass flow sensors.

allow the farmers to gather landscape imaging in real time with smooth workflow of processing and data storage. Moreover, IoT-based systems are effective in monitoring agricultural environmental parameters such as plant growth morphology, air temperature, light intensity, soil moisture, humidity, CO_2 concentration, pH values, and crop water use efficiency. In a study, CC2430, an intelligent irrigation system was developed and employed to evaluate multiple agricultural parameters. Notably, cost for implementation of the system was 44.8% less compared to similar foreign products and the crop water use efficiency was 22.6% compared to traditional irrigation methods [51]. Interestingly, smart phones, smart watches, and laptops can be coupled with UAS to gather visual and environmental information in more convenient ways. Multipoint assessment system with field surveillance, data acquisition, and monitoring are accurately performed using IoT based devices. In green house set-ups, IoT technologies facilitates controlled supervision and computerized data processing that have strengthened the applicability of the digital system in pest and disease management. IoT-enabled devices facilitate farmers to trace pests in real time to take proper preventive measure in a short time frame. Different sensors like DHT11, LDR, DS18B20, Soil Moisture sensors, Noir camera, single-board microcontrollers, and Application Programming Interfaces (APIs) help gather data associated with crop growth parameters such as intensity of sun light, etc. [52]. Smart Crop Selection (SCS) model utilizing ML algorithms such as Decision tree, SVM, KNN, Random Forest, and Gaussian Naïve Bayes is based on data of metrological and soil factors including nitrogen, phosphorus, potassium, CO_2, pH, temperature, humidity of soil, and rainfall. This model can provide prediction of the rainfall, a much-desired information for the health of any crop with 97-98% accuracy [53]. This will enable the farmers to adopt necessary measures to enhance crop growth.

In 2019, a performance evaluation of Smart IoT-based irrigation monitoring and control system was conducted by Abba et al. [54] and the result was found satisfactory. Loamy and sandy soils were selected as they were the prominent types of soil that are suitable for autonomous irrigation farming. The system counter was set to 0 to 300, and various moisture readings were recorded with different counter values. It was observed that, with the inflowing irrigation to the soil, the soil tended to absorb the water slowly. Interestingly, at a particular point it also retained water, causing the water pump to steadily switch off. This demonstrated that, the smart IoT-based irrigation monitoring and

control system can perform satisfactorily when engaged in irrigating farmland of loamy soil. Furthermore, the electricity expenses were reduced by shutting down the water pump when the moisture level exceeded 400% [54].

3.2. Application of IoT on weather sensing

IoT has become popular to sense various parameters of weather like humidity, temperature, and soil moisture. IoT with wireless network and smart sensors provides real time information on environmental parameters to carry more précised farming in vineyard and winery [55]. If the essential environmental factors are changed beyond the threshold level, IoT-coupled with sensors automatically sends alarm to the administrator to take necessary action against the upcoming adversities. The smart system also has surveillance on CO_2 content, luminance, humidity, and temperature for crop growth in real time [56]. Deployment of an IoT-enabled ecosystem, which integrates smart sensors and devices to exchange data, is the latest technology for weather surveillance and improved decision making in agro-practices [57].

Interestingly, IoT with smart sensors are gaining popularity in cucumber farming to sense cryogenic disasters with environmental parameters in real time basis. In this method, IoT-coupled sensors sense temperature and sends an alert massage to the administrator when the temperature falls below the threshold level. This allow administrator to activate heating equipments to save plants from cold shock. IoT can also be combined with third party weather forecasting services to receive updates on the upcoming changes in weather conditions to take appropriate measures against unfavorable changes in climate [58]. IoT sensor nodes coupled with MATLAB Simulink having flexible architecture and self-tuning reference input facilities enable growers to monitor microclimates and gain a better insight into interpreting crop growth environment in green houses [59]. Custom-built hardware tagged with IoT based software viz. Low-Power Wide-Area Network (LoRaWAN) transmitter nodes, a multi-channel LoRaWAN gateway, and a web-based data monitoring dashboard can be used to collect microclimate data. Multiple MATLAB Simulink blocks are used to process collected data to detect modulation in microclimate parameters. Therefore, IoT sensor node and the Simulink model effectively provide farmers with a better insight into interpreting crop growth environment [60]. Multi-tier cloud-based IoT platforms containing sensors, controllers, and actuators enable farmers to monitor, control, and manage weather conditions and crop growth from long distance [61].

3.3. IoT based monitoring of soil properties

IoT coupled technologies with smart sensors can monitor various aspects of farm soil that are critical to plant growth and development. IoT can sense soil moisture, temperature, and nutrient contents and these information can be provided to the farmers [62]. These data can be calibrated and remotely accessed by farmers to take sufficient precaution measures against the crop diseases and pest attack. Therefore, it could be helpful to the farmers to carry smart farming [58]. IoT also enables farmers to track soil pH and rhizosphere zone multiparameter from remote area and to take preventive measures [51]. Plant growth and development in various soil types can also be monitored [51]. Therefore, smart IoT system provides farm workers with current updates on temperature, soil condition, and moisture to improve overall crop production and crop quality.

4. Sensors for smart farming

Multiple sensors have emerging implications in smart farming (Table 1). These sensors could aid in automated harvesting of crops, monitoring of environmental conditions, and measurement of crop

Table 1
Smart sensors and their utility in agriculture.

Different types of sensors	Utility	Working procedure	References
Acoustic based sensors	Detection and monitoring of pest population; harvesting of fruits.	Measures alterations in noise level in agricultural fields	[66]
Electromagnetic sensors	Record electromagnetic responses; measures residual nitrate level and organic matter concentrations in soil; real time measurement of transpiration rate.	Employs electrical circuits that record conduction or accumulation of electrical impulses in soil	[69,121]
Light detection and ranging (LIDAR)	Agricultural land-based utilities such as 3D modeling; soil erosion monitoring; agricultural land mapping; soil type detection.	LIDAR sends light wave pulses to the target object. After colliding with the target object, light wave pulses return back to the sensor. The time taken by light waves to return back to the sensors after colliding with the target object is considered for assessment.	[122]
Optical sensors	Employed to sense soil parameters like soil texture; mineral contents; clay content; moisture and color of soil. Fluorescence-mediated optical sensors are used to sense fruit maturation. Integral optical sensors along with microwave scattering is used to monitor orchard canopies.	Changes in light reflectance is assessed	[121,123,124]
Mechanical sensors	Measures soil mechanical resistance and soil particle compactness.	Sensors measure the force assessed by load cells or strain gauges.	[92]
Mass flow sensors	Assist in assessing yield production by measuring the grain flow through a combined harvester.	Sensors record the mass flow of grains through various modules such as moisture content sensing sensor; data storage, as well as an internal system.	[92]
FPGA (Field programmable gate array) based sensors	Measures moisture content; humidity; transpiration rate of plants; irrigation	Employs digital circuit surrounding silicon-based chips and logic blocks	[125,126]
Electrochemical sensors	Helps in measurement of nutrient status and pH of soil.	Individual sensors recording electrochemical gradients in agriculture soil.	[127,128]
Eddy covariance-based sensors	Record changes in levels of various gases including green house gases such as CO_2, methane, water vapor etc in agricultural lands.	Measures continuous over large agricultural lands.	[129]
Airflow sensors	Assess soil-air content and permeability; moisture content and in mobile or static conditions.	It senses several soil properties using unique identifying characteristics.	[130]
Ultrasonic ranging sensors	Helps in pest detection; crop canopy monitoring; weed recognition.	Employs ultrasonic sensor that sends and receives ultrasonic pulses to detect proximity of the object.	[131]
Flexible and wearable sensors	Helps in sensing shape, size, and growth of different parts of the plants, It also help sense temperature.	Sensors made up of stretchable materials so that they can be mounted on plant parts.	[95,96]
Battery-free and self powered sensors	Help in sensing environmental factors such as temperature, humidity etc. They monitor quality of food products.	Sensors are powered by solar cells and therefore do not rely on application of batteries.	[100,102]

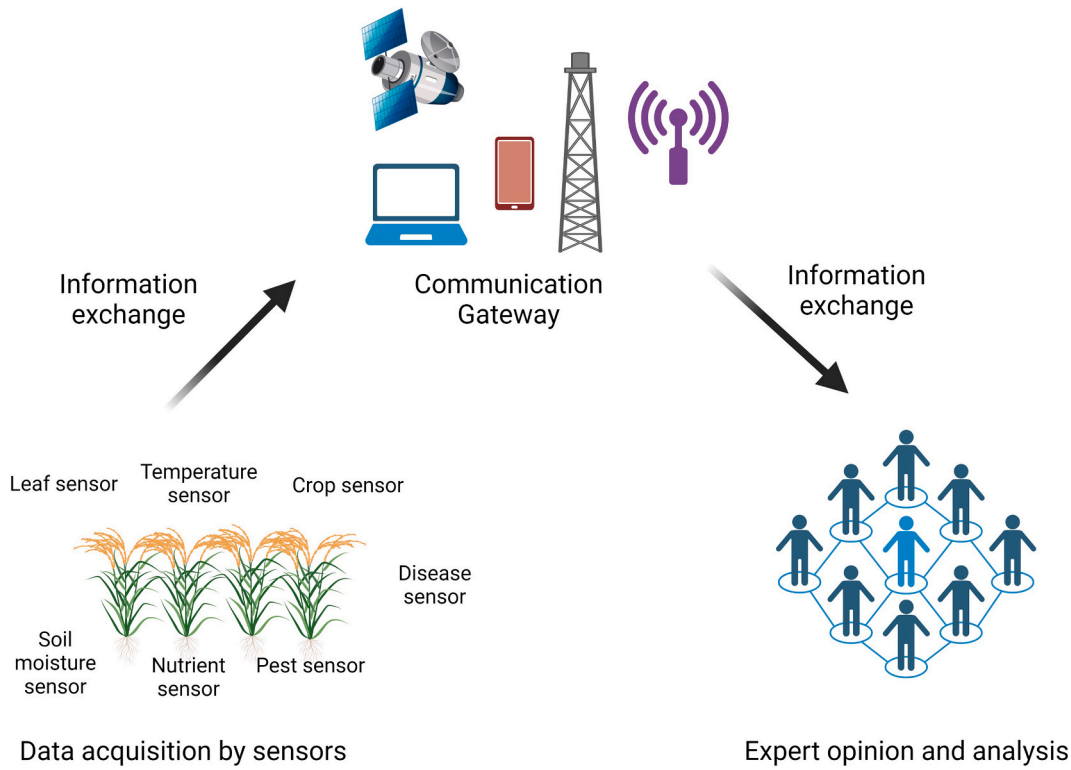

Fig. 2. Integration of IoT devices, farms, and human networks. IoT based sensors employed in farm lands sense a variety of crucial factors such as temperature, humidity, nutrient contents, etc., and send the data to communication gateway such as cloud servers, satellites, and computing system. These IoT-enabled devices are then used by human networks for analysis, decision making, and further required actions on farms.

yields. Sensors integrated with chips designated as smart sensors. Smart sensors can automatically record several environmental data and other related information in agricultural set ups with higher accuracy and store them in drives. These data are processed through microprocessors for interpretation and analysis. Smart sensors are the integral components of IoTs and data are transmitted using internet. It comprises wireless and actuator network system involving few to thousands of nodes connected to sensor hubs. Therefore a smart sensor is constituted of a sensor device, microprocessors, and wireless communication technology that enable remote surveillance of several factors in agriculture. Smart sensors can be coupled with other components such as amplifiers, transducers, analog to digital converters, and analog filters to enhance their performance (Fig. 2). Some useful sensors have been discussed below.

4.1. Acoustic based sensors

Acoustic based sensors are the electronic devices that are used to detect changes in sound frequencies. These sensors have the ability to sense vibrations that are returned from a moving object. Various insects generate sounds during general movement, feeding, and mating events. Such sounds can be detected by the various acoustic devices such as portable accelerometer and microphone systems [63]. Data gathered through the acoustic devices are transferred to the global databases and can be analyzed by world wide researchers for further development of automated acoustic sensors for more accurate pest detection at species levels [64]. During agricultural practices, acoustic sensors may be covered with dust and soil particles; however such coverings do not alter their working efficiency and the sensors continue to provide more accurate information regarding the target objects. The major advantage of using acoustic sensors is their higher sensitivity, low cost maintenance, and greater accuracy. Acoustic based sensors are typically used to detect and monitor various pest populations in agricultural lands. Additionally, estimation of the water volume in water sources is an essential factor for scheduling irrigation in automated irrigation systems. Ultrasound flow meters, which use the IoT-enabled ultrasound distance sensors, are now gradually replacing the mechanical flow meters. It is feasible to use underwater ultrasonic scanning to monitor the growth of aquatic plants, which has enormous potential for determining the harvest stage and growth rate [65]. Moreover, it also plays critical role in harvest of fruits [66]. An intriguing application of acoustics in agriculture is the estimation of fruit ripeness. Daosawang et al. [67] have claimed that, a sound generator and receiver may be used to determine the ripeness of watermelons.

4.2. Electromagnetic sensors

Electromagnetic sensors are useful in detecting agricultural soil and contamination. It also assist in mapping of topological features of agricultural lands. Electromagnetic sensors detect wide spectrum of electromagnetic waves that are received from various objects. The technique employs electrical circuits that record conduction or accumulation of electrical impulses in soil. Usually, sensing techniques and equipments such as electromagnetic induction, ground-penetrating radar, and gamma radiometric devices are deployed through unmanned flying vehicles to gather information of high resolution insights on surface processes and vegetation types. Ground penetrating radar and electric magnetic induction have been implicated in estimation of soil-water content [68]. Moreover, the sensors are useful to measure residual nitrate level, organic matter concentrations in soil, and real-time measurement of transpiration rate [69]. Ground-penetrating radar (GPR) and electromagnetic induction (EMI) techniques can be used to evaluate soil properties, states, processes, and their spatiotemporal variability. Interestingly, GPR and EMI techniques offer advantages due to their high resolution, ability to take non-destructive repeated measurements, man-power saving potential, and having broad spatial coverage with

geo-referenced data within agricultural landscapes [70]. In a study, electromagnetic induction sensor, EM38 was employed to draw relationship between electrical conductivity of soil and some soil properties like cation exchange capacity, soil moisture content, soil clay content, field capacity, etc. Results indicated significant similarity between electrical conductivity readings of soil computed from laboratory and sensor [71]. The findings advocate that, electromagnetic sensors could be helpful in accurately determining various physico-chemical parameters of soil in agricultural fields.

4.3. Electrochemical sensors

Electrochemical sensors are the light weight eco-friendly sensors that are easily portable to the agricultural lands. These sensors have efficiency to monitor plant growth, diseases, and environmental pollution more accurately in real time. Moreover environmental impacts of electrochemical sensors are comparatively less. Several sensors have been tested for biochemical parameters that are critical to agricultural yields. For instance, humidity sensors containing graphene oxide is employed in sensing plant leaf water [15]. NO_2 sensor consisting silver and reduced graphene oxide can be used to sense toxic gases in the agricultural fields. For example, a fully flexible device was constructed using metallic single-walled carbon nanotubes as conductive electrodes and AgNPs-decorated reduced graphene oxide as sensing layers. The device was able to sense as low as 0.2 ppm NO_2 at room temperature. Moreover, sensitivity was about 3.3 and 13 times greater than that of the sample based on metal electrodes and the sample without AgNP decoration respectively [72]. Certain sensors are useful in detecting volatile organic compounds in soil. Such sensors typically made up of carbon nanotubes, graphite, reduced graphene oxide, and gold nanoparticles. An imperceptible sensor patch integrating an array of graphene-based sensing materials and flexible silver nanowire electrodes on a kirigami-inspired stretchable substrate was developed and mounted on live tomato plants to profile key plant volatiles at low ppm concentrations with escalated response. Interestingly, multiplexed sensor array enabled detection of thirteen individual plant volatiles with >97% classification accuracy [73]. Some of the electrochemical sensors have been devised to detect plant sap and nitrogen in soil [74]. Therefore, electrochemical sensors have immense potential to sense wide range of parameters in agricultural lands that might be useful to enhance agricultural production (Table 2).

4.4. Light Detection and Ranging (LIDAR)

LIDAR technology is based on the light-based measurement of distance between sensor and a target object. Since light travels in a constant speed, LIDAR can precisely estimate the distance between the sensor that emits the light and the object to which the emitted light collide. LIDAR periodically emits light pulses and detects a series of collisions to built map of surrounding environment. Advanced LIDAR system is equipped with lesser pulses that enhance the efficacy of the equipment. Other performance features include operating range, the estimated error, and the scanning frequency. Coordinates obtained through light collision are stored in a file known as "cloud point" that helps to build 3D space. LIDAR technology has been used in variety of crop-related applications. For instance, it is used to monitor fruit ripening and vegetation structure. In addition, LIDAR sensors are used to measure agricultural landscaping and topography. Structural characteristics of trees such as leaf area index and canopy volume are sensed using LIDAR technology. The emerging technology is also employed for crop biomass estimation, phenotype characterization, and crop growth [75]. Interestingly remotely sensed LIDAR data are recently being used to derive high-resolution digital elevation models (DEM) to investigate field systems at a scale not previously possible. Moreover, integration of slope contrast mapping can be used to spot complexes of agricultural terraces [76]. Implication of FX6 LIDAR to sense Nippon Signal could be promising in detection and segmentation of plants and ground [77]. It is an indispensable prerequisite to perform localization, mapping, and navigation for autonomous agricultural robots. Therefore, LIDAR could have great potential in mapping agricultural fields and in initializing automated agriculture practices.

4.5. Optical sensors

Wide range of optical sensors has potential implications in agriculture. Such sensors are based on their ability to detect lights of various wavelengths. Light of specific wavelength is emitted from a light source to collide with the target object. The reflected light is sensed by the optical sensor to generate reflectant data and stored as text file. Freidenreich et al. [78] employed non-distructive optical sensors such as SPAD and GreenSeeker™ for analysis of nutrient uptake. Parameters like Normalized Difference Vegetation Index, soil leachate, and total leaf Carbon:Nitrogen levels were examined. Authors suggested that application of optical sensors is promising in determining fertilizer requirements by plants. Proximal optical sensors such as SPAD-502, GreenSeeker, and Canopeo App was used and found effective in correct prediction of crop biomass at harvest and monitoring plant's canopy expansion and senescence. Here, in-season predictions of crop biomass at harvest by SPAD-502 and GreenSeeker optical sensors were successful in terms of the coefficient of determination as 0.68 and 0.82, respectively. Moreover, the relative error percentage demonstrated significantly greater performance between observed and predicted values of 5.80% and 4.12% for GreenSeeker and SPAD-502, respectively [79]. Optical coherence tomography has been reported for it's potential implication in monitoring seed germination [80]. GreenSeeker™ optical sensor is helpful in managing nitrogen-based fertilizers during planting of wheat for better yield [81]. Information on weed distribution in agricultural fields has been successfully gathered using optical sensing technology. In a study using a tolerance threshold, optical sensor coupled with spectroscopy was observed to be highly accurate in detecting cells infected with green weeds. An overall percentage agreement of 90.9% was recorded and value was comparable with 90.5% and 91.2% of two reference methods used [82].

4.6. FPGA (Field Programmable Gate Array) based sensors

FPGA are integrated circuits with configurable interconnections and logics. These work with the one or more microprocessor cores, cache, memory, and reconfigurable logics. As per the need, these can be reprogrammed for multiple times in hardware/software systems. FPGA

Table 2
Different types of electrochemical sensors and their applications.

Types of electrochemical sensors	Sensing materials	Applications	References
Volatile organic compound sensor	Carbon nanotubes/Graphite	Plant health monitoring	[73]
Volatile organic compound sensor	Reduced graphine oxide/gold nanoparticles	Plant health monitoring	[132]
Humidity sensor	Graphine oxide	Leaf water monitoring	[15]
Nutrient sensor	Reduced graphine oxide/Ion selective electrode	Plant sap sensing	[74]
Pesticide sensor	Lase-induced graphene/Gold nanoparticles	Assessment of pesticide contents	[133]
O_3 sensor	Poly (3,4-ethylene dioxythiophene)	O_3-mediated damage monitoring in plants	[74]
NO_2 sensor	Silver/reduced graphine oxide	Harmful gas detection in agricultural lands	[72]

with fuzzy logic has been tested in ground level and was found effective in providing future productivity and recommendations. In this study, an intelligent embedded fuzzy decision support system (IEFDSS) was tested at ground level and accuracy of obtained results was 96% higher than those of the present methods [83]. FPGA can also be configured for use in a system to monitor agricultural environment. Such monitoring system is equipped with different sensors, wireless protocol, screen, microcontroller unit, field-based FPGA, and serial protocol. Critical environmental factors such as temperature, humidity, and soil moisture are sensed by respective sensors and fed to the FGPA to display the information in the screen [84]. In a study, FGPA-based image processing (FIP) device was used and found effective in eliminating the technical issues related to image resolution and processing speed of the conventional imaging systems. Here, the FIP system was reported to depict high accuracy with a Lin's concordance correlation coefficient of 0.99 and 0.91 for the DLSR and web camera reference system, respectively [85]. Therefore is it obvious to state that, FIP system has better accuracy over web camera reference system. An automated system to control irrigation was also successful by programming VHSIC Hardware Description Language on FGPA to display and measure humidity [86].

4.7. Eddy covariance (EC) based sensors

EC based sensors are employed to record latent heat and sensible heat fluxes to measure the evapotranspiration and use of water by crops. EC is also useful in measuring the green house gases. It measures the vertical concentration gradients of Carbon dioxide, Methane, and other gases. EC can be coupled with remote sensing models to better estimate green house gas balance as well as gross and net primary productivity of crops [87]. Quantum cascade laser-based EC measures the surface-atmosphere exchange of Ammonia [88]. In a field based study, evapotranspiration along with soil heat flux was measured utilizing EC during sprinkler irrigation. Results indicated that, for mature cotton of height approaching 1 m, nondimensional evapotranspiration ranged from 1.4 to 1.6 times the reference (non-irrigating) evapotranspiration value [89]. In another investigation, Liang et al. [90] measured Nitrous oxide in different seasons and environmental conditions using EC. They suggested that EC could serve as an alternative in national Nitrous oxide inventories. In another similar study, CO_2 fluxes was measured using EC and found that pasture system regulates the atmospheric CO_2 balance [91].

4.8. Mechanical and mass flow sensors

Physical and chemical factors of soil are responsible for variability in crop yield across the agricultural lands. Soil strength reflects the compactness of soil particles in a particular area. Higher soil compactness imposes negative impacts on crop yield. Mechanical sensors can be used to measure the soil compactness and mechanical resistance in soil. This information could be helpful to customize the irrigation process according to the mechanical features of the land. Honeywell FSG15N1A is an example of mechanical sensor. Honeywell FSG15N1A senses the force that is exerted by plant roots to absorb water. In a study, vertical and horizontal sensors were used to measure the soil compactness in a given area. This integrated approach utilized an ultrasonic distance sensor to measure the depth penetrated by coulter into the topsoil layer. Results indicated that, the integrated approach was useful to sense the alterations in top soil strength according to the depth. This could help in mapping the mechanical resistance of soil in agricultural lands [92].

Mass flow sensors provide scopes for automated monitoring of crop yield. In a study, the accuracy of mass flow sensors was tested using a portable grain mass flow test rig. Analysis revealed that, the slope of grain elevator can affect the calibration of sensors and therefore has potential to alter the accuracy of the system. The calibration was altered at 10° pitch that significantly modulated the accuracy of the mass flow sensor. Of note, the accuracy of the mass flow sensor was greatly appreciable as the coefficient of determination was recorded as 0.99 if the device was run with calibrated flow rate [93]. The impact based grain flow sensors are widely used to compare the agricultural yields. Nonetheless, it may produce false signals due to excessive natural vibrations. To enhance the accuracy of the sensors, a dynamic compensation algorithm could be helpful to minimize the overshoot and natural vibration signals [94].

4.9. Flexible and wearable sensors

Flexible sensors are gaining popularity due to their ability to provide real time physiological data of plants. These sensors can be directly deployed to plant surfaces for better detection of signals. Flexible sensors are designed by considering the complex surface of plant parts. Leaf sensors are made up of porous substances that do not interfere with the respiration of plants through leaf. Other sensors include the flexible fruit sensors that can easily realize a planer surface to the three dimensional architecture and therefore it can work on variable surfaces without any loss in working efficiency. These sensors are designed to record data of temperature changes at microenvironment level [95]. The stretchable flexible sensor made by mixing graphite powder and chitosan solution in certain proportions to measure growth of fruits. For protection from humidity, these sensors are covered with rubber. These sensors were directly written on cucumber fruits to measure their elongation and the sensors can reach a strain up to 60% [96].

Wearable sensors are mainly developed to monitor human health. However application of these sensors in agriculture is still in infancy. Wearable sensors can directly be placed on various plant parts like stems, leaves, etc., to study changes in microenvironment parameters such as temperature which can be used to detect diseased conditions and several biotic and abiotic stress on plants. Some stretchable sensors made up of polydimethylsiloxane stretching material (PDMS) layered by Ti/Au metal film have been employed in measuring stem elongation. Such sensors have shown a linear detection range of 0% to 22% strain, corresponding to an elongation range of 0 to 3.75 mm in barley stem [97]. Wearable sensors made of stretchable substance latex, coated with graphite ink and carbon nanotube ink was placed on fruits of *Cucurbita pepo* to monitor circumferential elongation. This sensor has efficacy to measure elongation at micrometer scale. Notably, this approach encompasses simplified procedure, low implementation cost, enhanced speed, and nanoscale sensitivity that suggest promising scopes in accurate measurement of plant growth [98]. Wearable sensors made up of Gallium-based liquid alloy with high fluidity and electrical conductivity has the stretchability up to 200%. These sensors can adopt the irregular shapes and have been tested on roses and bean sprout [99].

4.10. Battery-free and self powered sensors

Battery-free and self powered sensors are advantageous over the conventional sensors because they eliminate the maintenance of batteries on a regular basic and reduce the wastage of time caused by battery replacement. Therefore conventional rechargeable battery-based sensors pose a challenge for long term application. Therefore, these sensors are often considered as install-and-forget sensors and enable the farmers to monitor plant health and growth parameters throughout the cultivated area. Some battery free sensors employ the platform equipped with off-the-shelf components on a printed circuit board (PCB) that can harvest and supply energy to sensors and have facilities of Bluetooth Low Energy communication [100]. A battery-free wireless moisture sensor system (BWMS) coupled with Near Field Communication (NFC) reader was devised to monitor self-life of fruits. BWMS can acquire weight and temperature sensor data in real time from fruits and predict self-life to indicate quality of the fruit. In a study involving micro-irrigation system, small solar powered cell was designed to power sensors for real time data acquisition. The technique was economical and suggests an opportunity for electrification in agricultural sectors

[101]. Solar cells were used to power supply WSN-based ZigBee (CC2420 module) wireless transceiver and microcontroller (low-power ATmega128L) to monitor temperature, leaf wetness, humidity, and rain level in vineyards [102]. In another study, solar cells were used to supply electricity to WSN employing low power microcontroller and ultra low power transceiver to monitor changes in humidity and temperature in a green house [103]. Battery-free Radio Frequency IDentification (RFID) sensors have also been developed to detect food freshness and microbial growth with high selectivity and sensitivity [104].

5. Pest monitoring in smart farming

Some devices capable of producing ultrasonic sound waves can repel a number of pests in agricultural fields. For instance, Tiwari et al. [105] devised an electronic pest repellant that produces strong ultrasonic sound waves. These waves repel pests like rodents, insects, and others. Therefore, such devices are good alternative to chemical pesticides that impose numerous adverse impacts on beneficial organisms [106–112]. Moreover, unintentional exposure to chemical pest repellents can also contribute to enhanced usage of ultrasonic equipments. Electronic pest repellants are cheap, environmental friendly, and impose no adverse impact on human. Mainly such devices target the hearing senses of pests to repel them.

Certain image processing algorithms are extremely helpful in detecting pests. Spectral camera technology with the help of UAV can capture the high resolution images of the farm area at macro and micro levels. Such images can be analyzed to detect pest infestation and onset of associated diseases in crops [113]. An integrated approach combining Parallel and Distributed Simulation Framework (PDSF) with IoT can improve the pest monitoring in agricultural fields. This integrated system will reduce the workload on a single GPU and can distribute the workload to other available GPUs and multiple cores. Such system can improve the overall performance to better monitor pest population. Some camera based automated traps can be implemented in controlling pest population. Guarnieri et al. [114] modified a commercially used trap into an automated trap having potential of data acquisition and data processing to control codling moth *Cydia pomonella* L in apple orchards. Ünlü et al. [115] formulated a delta-shaped trap with a high resolution camera, solar panel, battery, a charging unit, and a GPRS. This automated trap was used to european grapevine moth *Lobesia botrana* Denis in vineyards. Sticky delta traps was designed by Shaked et al. [116] to monitor Mediterranean fruit fly *Ceratitis capitata*. Bucket traps integrated with camera-based electronic McPhail trap was used to trap fruit flies *Bactrocera oleae*. Real-time images of the trapped fruit flies were automatically captured and sent to a server for storage. Using the captured photographs, entomologist were able to identify the fruit flies with >88% accuracy. A multi-funnel trap integrated with camera system is effective in remote monitoring of forest longhorn beetles and bark beetles [117]. Examining insect behavior by the implication of multiple wireless sensors could also aid in more accurate detection and characterization of insect pests in agricultural fields [118] (Fig. 3).

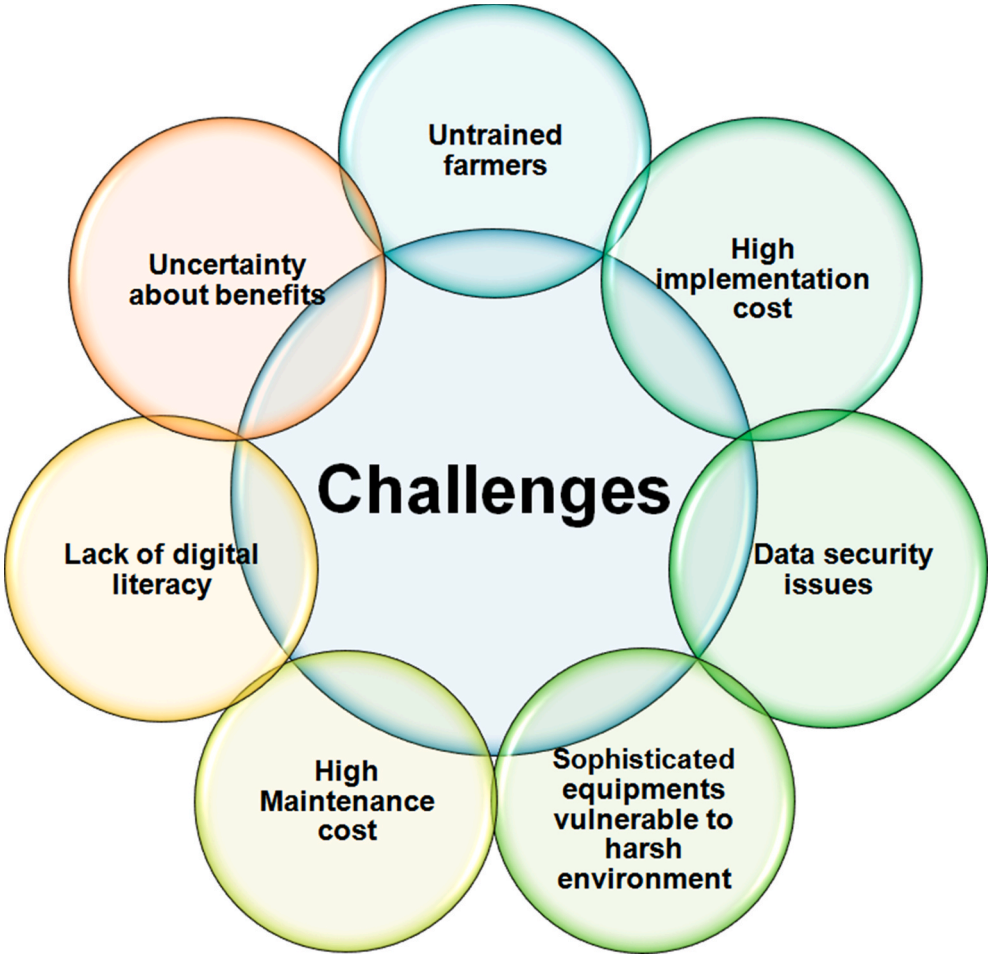

Fig. 3. Different challenges encountered with regard to the deployment of IoT-based smart sensors and accessory devices for fully automated farming.

6. Challenges and future possibilities

Global studies on IoT and smart sensor-based smart farming have produced satisfactory results. IoT-based technology has been implemented in multiple small scale agricultural fields. However, deployment of the same on a large scale level is still pending. One of the major challenges is the financial cost that will be imposed during the deployment and installation of the IoT-tagged sensors and accessories in a large area of agricultural land. Moreover, there is uncertainity between the cost of implementation and the amount of profit gained. Major cost for implementation of the IoT-enabled technology includes the cost for hardware purchase, software installation, and system operation. Other additional costs for the same will be applicable for the energy consumption, system maintenance, service registration, and cost for man power involved in operating the integrated hardware devices and associated software. A major drive with respect to the technology-based literacy of farmers is required on a global scale for worldwide application of the IoTs. Lack of proper knowledge and literacy regarding the scopes, mode of action, and implementation of IoT-based technology in faming can result in underutilization of the smart system in agriculture. Deployment of IoT-coupled smart sensors and accessories in rural farm areas, where the farmers are less familiar to advanced technologies could be more challenging [35]. Government policy makers are required to formulate economic policies for the successful and better productive implementation of IoT by farmers in farmlands.

Data privacy and security is addition problem that can negatively impact the implementation of IoT and smart systems on a large scale. Data stored in cloud servers can be tempered by attackers to damage the automated farming processes in farm lands. Such attacks could further affect the productivity of the entire farm land and improper management of environmental factors. Therefore, issues related to data security of IoT are among the major factors that are responsible for slow adoption of technologies for smart farming [119]. Smart farming requires a reliable encryption system for protection of important data and digital systems from the global cyber attackers. Cryptography integrated with the strong keys might limit the cyber attacks on cloud servers. An additional method can also provide a reliable outcome where multiparty computation system integrated with homomorphic encryption or coupled with block chain method are implemented [120].

One of the major problems with the IoT devices is that they are installed outdoor and hence these devices are exposed to harsh environmental conditions like heavy rainfall, dust, wind, temperature, etc. Such adverse environmental conditions might result in unexpected mechanical failure of the sophisticated devices. Hence, manufacturers of the IoT devices for smart farming should use raw materials that could withstand with such adverse environmental factors therefore contributing to the longevity of the sophisticated devices with more reliable output.

7. Conclusion and Outlook

IoT-enabled smart farming has great perspectives in future. Advanced sensors can be employed to monitor environmental factors such as rainfall, moisture, and temperature for maximum agricultural yield. IoT-enabled devices could help measure water and Nitrogen contents in soil. In addition, based on CO_2 level in farm lands, evapotranspiration rates can be effectively monitored to have better surveillance on crop health. Moreover, pest attack can be reduced by controlling pest population through IoT-enabled traps equipped with high resolution camera and other accessories. Despite such advantages, major challenges for the deployment of IoT-based smart sensors in agriculture on a larger scale include purchase and maintenance cost of sophisticated hardwares and softwares. In addition, farmers residing in rural areas lack sufficient knowledge regarding the use of IoT devices. Finally, cyber attackers might influence the automated smart farming by damaging the cloud servers where important data are stored. Government policies for financial support to farmers, digital literacy, and data encryption could ease the implementation of smart farming on a large scale.

CRediT author statement

Prem Rajak: Conceptualization, Investigation, Data Curation, Writing- Original draft preparation, Writing- Reviewing and Editing. Abhratanu Ganguly: Writing- Original draft preparation, Data Curation. Satadal Adhikary: Writing- Reviewing and Editing. Suchandra Bhattacharya: Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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